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[23TCSE307]

THIRD SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	List the types of Machine learning.	L2	CO1	**02**
	b.	Describe Feature Engineering.	L1	CO2	**02**
	c.	Define Decision Tree.	L2	CO3	**02**
	d.	How to minimize the error in back backpropagation algorithm?	L4	CO4	**02**
	e.	What are the ethical considerations in machine learning?	L3	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Outline the concept of unsupervised learning in the context of ML.	L2	CO1	**05**
		OR			
	b.	What are the fundamental mathematical concepts that form the foundation of machine learning and how do these concepts contribute to the development and understanding of machine learning algorithms?	L3	CO1	**05**
3.	**a.**	How does the filter method for feature selection work? What are some common statistical techniques used in this approach, such as correlation coefficients or chi-square tests?	L4	CO2	**05**
		OR			
	b.	What is data visualization, and how does it facilitate the understanding and interpretation of complex datasets?	L3	CO2	**05**
4.	**a.**	Explain the concept of non-linear decision boundaries in SVMs and how kernels help to achieve them.	L3	CO3	**05**
		OR			
	b.	What are the main advantages of using Principal Component Analysis(PCA) for dimensionality reduction and also list challenges or limitations associated with PCA.	L1	CO3	**05**
5.	**a.**	How do RNNs differ from traditional feedforward neural networks.	L4	CO4	**05**
		OR			

	b.	What are the key metrics used to evaluate classification models? Describe them.	L2	CO4	**05**
6.	**a.**	How can you read from and write to various file formats (CSV, Excel, SQL) using Pandas?	L3	CO5	**05**
		OR			
	b.	How do different types of adversarial attacks, such as evasion attacks, poisoning attacks, and inference attacks, impact the security and reliability of machine learning systems?	L3	CO5	**05**
7.	**a.**	Differentiate between supervised learning and unsupervised learning?	L2	CO1	**05**
		OR			
	b.	Summarize the concept of convolutional neural networks in Deep learning?	L1	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss in detail about different types of techniques used in machine learning.	L1	CO1	**10**
		OR			
	b.	Write short notes on **i.** Gradient descent and **ii.** Optimization methods	L2	CO1	**10**
9.	**a.**	What are the essential techniques for cleaning and preparing data to ensure it is suitable for training machine learning models, and how do these techniques impact model performance?	L4	CO2	**10**
		OR			
	b.	What are some methods for handling missing data during the EDA phase, and how can these methods affect subsequent analysis?	L3	CO2	**10**
10.	**a.**	Define clustering. Show the types of clustering techniques and explain any one of the clustering techniques.	L2	CO3	**10**
		OR			
	b.	Irradiate the concept of Reinforcement Learning and how these concepts are applied in real-world scenarios?	L4	CO3	**10**

11.	**a.**	Explain the concept of different layers in Neural network? Describe each one in brief.	L2	CO4	**10**
		OR			
	b.	Enumerate techniques for deploying ML models in real-world applications.	L4	CO4	**10**
12.	**a.**	How can you create different types of plots (e.g., line plots, scatter plots, histograms) using Matplotlib?	L6	CO5	**10**
		OR			
	b.	How can fairness be defined and measured in the context of machine learning, and what metrics are commonly used to assess it?	L3	CO5	**10**
13.	**a.**	Compare and contrast Linear Regression and Logistic Regression in supervised learning.	L4	CO3	**10**
		OR			
	b.	What are the fundamental concepts of artificial neural networks, including their architecture, training process, and key components, and how do these concepts contribute to their ability to model complex patterns in data?	L4	CO4	**10**